Simplify the following expressions.

1. $\dfrac{3x^2-12}{x^2-4x+4}$

Solution:

$$\frac{3x^2-12}{x^2-4x+4} \ggg \frac{3(x^2-4)}{(x-2)^2} \ggg \frac{3(x-2)(x+2)}{(x-2)^2}$$

$$\ggg \boxed{\frac{3(x+2)}{x-2}}$$

2. $\log\left(\dfrac{100}{x}\right)-\log(10)$

Solution:

$$\log\left(\frac{100}{x}\right)-\log(10) \ggg \log\left(\frac{100}{x\cdot 10}\right) \ggg \boxed{\log\left(\frac{10}{x}\right)}$$

4. $2^x\cdot 4^{x+1}$

Solution:

$$2^x\cdot 4^{x+1} \ggg 2^x\cdot(2^2)^{x+1} \ggg 2^x\cdot 2^{2x+2} \ggg \boxed{2^{3x+2}}$$

5. $\dfrac{x^4-16}{x^2-4}$

Solution:

$$\frac{x^4-16}{x^2-4} \ggg \frac{(x^2)^2-4^2}{x^2-4} \ggg \frac{(x^2-4)(x^2+4)}{x^2-4} \ggg \boxed{x^2+4}$$

6. $\dfrac{\frac{2}{x}+\frac{3}{y}}{\frac{1}{x}-\frac{1}{y}}$

Solution:

$$\frac{\frac{2}{x}+\frac{3}{y}}{\frac{1}{x}-\frac{1}{y}} \ggg \frac{\frac{2y+3x}{xy}}{\frac{y-x}{xy}} \ggg \boxed{\frac{2y+3x}{y-x}}$$

7. $\log_3(81)$

Solution:

$$\log_3(81) \ggg \log_3(3^4) \ggg \boxed{4}$$

8. $\dfrac{3^{2x}\cdot 9^x}{27^x}$

Solution:

$$\frac{3^{2x}\cdot 9^x}{27^x} \quad \ggg \quad \frac{3^{2x}\cdot (3^2)^x}{(3^3)^x} \quad \ggg \quad \frac{3^{2x}\cdot 3^{2x}}{3^{3x}} \quad \ggg \quad \frac{3^{4x}}{3^{3x}} \quad \ggg \quad 3^x$$

9. $\dfrac{(x^2-9x+20)(5^{x+2})(3x-6)}{(x-4)(x-2)(25)}$

Solution:

Simplify component factors:

$$\begin{aligned} (x^2-9x+20) &\quad \ggg \quad (x-4)(x-5) \\ 5^{x+2} &\quad \ggg \quad 5^x\cdot 5^2 \\ 3x-6 &\quad \ggg \quad 3(x-2) \\ 25 &\quad \ggg \quad 5^2 \end{aligned}$$

Therefore:

$$\frac{(x^2-9x+20)(5^{x+2})(3x-6)}{(x-4)(x-2)(25)} \quad \ggg \quad \frac{(x-4)(x-5)5^x\cdot 5^2\cdot 3(x-2)}{(x-4)(x-2)(5^2)} \quad \ggg \quad 3(x-5)5^x$$

10. $5^{2\log_5(7)}$

Solution:

First apply log power rule: $\quad 2\log_5(7) \quad \ggg \quad \log_5(7^2)$

Then:

$$5^{2\log_5(7)} \quad \ggg \quad 5^{\log_5(7^2)} \quad \ggg \quad 7^2 \quad \ggg \quad 49$$

11. $\dfrac{\log 216}{\log 6}$

Solution:

Notice that $216 = 6^3$. Then:

$$\frac{\log 216}{\log 6} \quad \ggg \quad \frac{\log 6^3}{\log 6} \quad \ggg \quad \frac{3\log 6}{\log 6} \quad \ggg \quad 3$$

Fun Fact:

$$\frac{\ln X}{\ln Y} = \frac{\log_b X}{\log_b Y} \quad \text{for any } X,\ Y,\ b>0$$

(Can you prove this? Hint: write $X = Y^{\log_Y(X)}$.)

For example:

$$\frac{\log e^3}{\log e^2} \quad \ggg \quad \frac{\ln e^3}{\ln e^2} \quad \ggg \quad \frac{3}{2}$$

Or similarly:

$$\frac{\ln 1000}{\ln 100} \quad \ggg \quad \frac{\log_{10}(1000)}{\log_{10}(100)} \quad \ggg \quad \frac{3}{2}$$

12. Is this equation necessarily true, assuming that a, $b > 0$?:

$$\cos\left(\cos^{-1}\left(\frac{a}{\sqrt{a^2+b^2}}\right)\right) = \frac{a}{\sqrt{a^2+b^2}}$$

Solution: Yes.

Recall that $\cos(\cos^{-1}(X)) = X$ provided $X \in [-1, 1]$. (The equation is false when X is not in that range!)

We know that $a^2 + b^2 > a^2$, so $\sqrt{a^2+b^2} > a$. Therefore the fraction $\frac{a}{\sqrt{a^2+b^2}}$ is in the range $(0, 1)$, thus definitely within the range $[-1, 1]$.